

Milin Patel

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PROFESSIONAL SUMMARY

AI/ML-focused Computer Science student specializing in Deep Learning, Generative AI, and intelligent system development. Experienced in building transformer-based research models, RAG-powered applications, and multimodal AI assistants using Python, PyTorch, LangChain, and modern AI frameworks. Demonstrated ability to develop scalable, real-world AI solutions across medical imaging, conversational AI, and productivity systems. Passionate about applying machine learning and data-driven technologies to solve impactful real-world problems through research-oriented engineering and practical implementation.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL
Tools: Flask, Gradio, HuggingFace, LangChain, Git, Streamlit
Concepts: Generative AI, Prompt Engineering, NLP, STT/TTS, RAG, REST APIs
Platforms: Linux, GitHub, Jupyter

PROJECTS

QABot with RAG and LangChain

IBM GenAI Capstone Project

- Engineered a contextual Q&A bot using LangChain and Streamlit, improving query response accuracy by **30%** over base models.
- Developed a RAG pipeline to retrieve and answer from custom document data, reducing hallucination by **40%**.
- GitHub:* github.com/Aestheticfrog/QAbotwithragandlangchain

C.H.A.O.S AI – Multimodal Voice Assistant

Oct 2024 - Present

- Developed a voice-based AI assistant using Flask, Gemini Pro API, STT/TTS, and Gradio, achieving **95%** task execution accuracy.
- Designed a modular architecture that improved feature integration time by **50%**. Used by **30+** peers in productivity trials.
- GitHub:* github.com/Aestheticfrog/CHAOSAII

Study Routine Coach – AI Productivity App

Jan 2025 – May 2025

- Developed a Flask + Gemini Pro-based study planner, resulting in a **20%** increase in task completion rates for early users.
- Integrated a robust REST API backend with prompt workflows, supporting over **100** API calls per minute.
- GitHub:* github.com/Aestheticfrog/study-gemini-schedule-coach

Alzheimer's Disease Detection using Deep Temporal Transformer

Aug 2025 – April 2026

Research Paper | PyTorch, ResNet34, Transformers, Medical Imaging

- Developed a Deep Temporal Transformer (DTT) model for Alzheimer's detection using MRI scan sequences with patient-level temporal learning.
- Integrated ResNet34 spatial feature extraction with Transformer-based temporal progression modeling, achieving an F1-score of 0.82.
- Implemented Brown Bear Optimization (BBO) for projection-dimension tuning and improved classification robustness on MRI datasets.
- Utilized PyTorch and patient-level stratified cross-validation to reduce data leakage and improve generalization.
- Research Paper: A Deep Temporal Transformer Approach for Alzheimer's Disease Detection
- GitHub:* github.com/Aestheticfrog/deep-temporal-transformer-alzheimers

EXPERIENCE & TECHNICAL LEADERSHIP

Data Science Intern

Celebal Technologies *Aug 2025 – Oct 2025*

- Did training as a new data science intern to get to know how a data science engineer think and his mindset. Improved my knowledge by almost **50%**.
- Developed predictive models to forecast customer churn, resulting in a **15%** improvement in retention rates.
- Utilized machine learning algorithms and statistical analysis to extract key insights, driving data-informed business decisions.

Collaborator

VITB AI Innovator Hub

- Selected for "Darzi: AI Resume Tailor Suite" project, which aims to automate resume formatting by **80%**.

Contributor

GirlScript Summer of Code (GSSoC)

- Actively contributed to an open-source project by resolving bugs and adding new features, improving code maintainability by **25%**.

Core Technical Member

Technomech Club, VIT Bhopal

- Coordinated a UI/UX hackathon with over **100** participants, onboarding mentors and judging submissions.

EDUCATION

VIT Bhopal University

B.Tech in Computer Science (AI & ML)

- Expected Graduation: 2027

CGPA: **8.58/10**

Bal Bharati Public School

Pre-University Education

- Completed: 2023

Class X : 96.7%

Class XII : 89.6%

CERTIFICATIONS

- IBM – Generative AI Professional Certificate (**16** courses completed)
- BoosCoder Academy – Advanced SQL for Interviews
- NPTEL – Cloud Computing

